

The cross-section $\mathbf{A} \rightarrow \mathbf{A}'$ (§1.544)

Freyd & Scedrov, §1.544 — the inflation and its section

A **cross-section** of the forgetful functor $\text{listProd} : \mathbf{A}' \rightarrow \mathbf{A}$ is a functor going the other way, $\mathbf{A} \rightarrow \mathbf{A}'$, that listProd undoes. The forgetful functor is **many-to-one**: it sends a list s to its product $\prod s$, and many different lists have the same product (up to isomorphism), so many lists sit over each object A . The cross-section chooses one of them — the one-element list $[A]$.

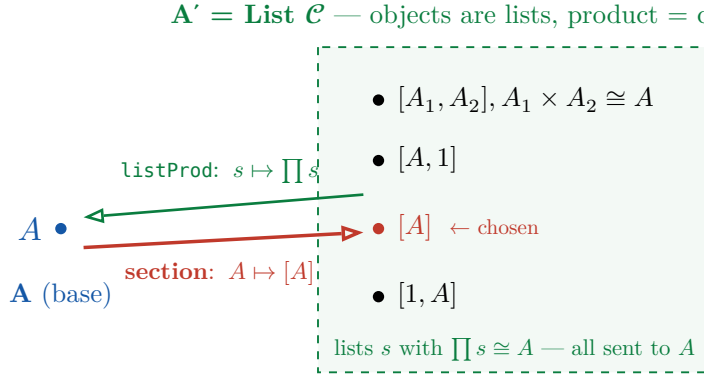


Figure 1: listProd (green) sends every list over A down to A (it is many-to-one). The cross-section (red) lifts A back up to the **single** list $[A]$. It is a **section** because going up then down returns to A : $A \mapsto [A] \mapsto \prod[A] = A$, i.e. $\text{listProd} \circ \text{section} = \text{id}$.

That round trip is the defining property — drawn as a triangle:

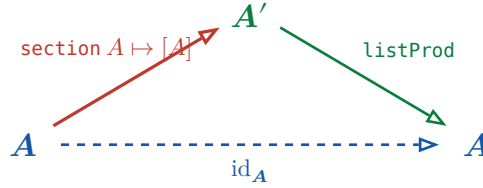


Figure 2: $\text{listProd} \circ \text{section} = \text{id}$: the cross-section is a one-sided (right) inverse of the forgetful functor.

Why §1.544 wants it. The slice inclusion $\mathbf{A} \rightarrow \mathbf{A}/B$ is split into two strict steps:

$$\mathbf{A} \xrightarrow{A \mapsto [A]} \mathbf{A}' \xrightarrow{s \mapsto s ++ [B]} \mathbf{A}'/B$$

The cross-section is the first arrow — it relabels each object A of $\mathbf{A}$ as the singleton list $[A]$, so the second step (appendFunctor B , “multiply by B ” done as list concatenation) has lists to append to.

§1. Δ separates objects

$\Delta : \mathbf{A} \rightarrow \mathbf{A}/B$ is the diagonal functor sending each object A to the pair $(A \times B, \text{snd} : A \times B \rightarrow B)$, an object over B . Recall the factorization $(- \times B) = \Sigma \circ \Delta$, where $\Sigma : \mathbf{A}/B \rightarrow \mathbf{A}$ is the forgetful functor.

“ **Δ separates objects**” means Δ is injective on objects: $\Delta(A) = \Delta(A') \implies A = A'$ (an equality of objects, not merely an isomorphism).

Why it holds under the §1.544 interpretation (the inflation $\mathbf{A}'$ whose product is list concatenation): the object $A \times B$ is the literal concatenated list, an equality-carrying object.

A slice object is its structure arrow, so $\Delta(A) = \Delta(A')$ forces equal domains $A \times B = A' \times B$ as lists, and lists cancel ($\mathbf{B}::s = \mathbf{B}::t \implies s = t$, the strict cancellation `strict_cancel = List.cons.inj`). Hence $A = A'$.

Why it fails in the ordinary interpretation (product only defined up to isomorphism): there you only ever get $A \times B \simeq A' \times B$, and an isomorphism never forces $A = A'$. Set example with $B = \mathbb{N}$: $\mathbb{N} \times \{*\} \simeq \mathbb{N} \simeq \mathbb{N} \times \mathbb{N}$, yet $\{*\} \neq \mathbb{N}$. So the plain diagonal is full and faithful but not injective on objects.

Why Freyd wants this: §1.544 wants $\mathbf{A}$ to be a genuine subcategory of $\mathbf{A}/B$, and a subcategory inclusion must be injective on objects. The inflation upgrades “ $\simeq$ (up to iso)” to “ $=$ (literally equal)”, converting the full-faithful Δ into an honest inclusion $\mathbf{A} \subset \mathbf{A}/B$.

§2. Δ separates morphisms (when B is well-supported)

Important distinction (this is the point of the section):

- $\Delta(f)$ is a morphism in the slice $\mathbf{A}/B$: the over-morphism $(\langle A \times B, \text{snd} \rangle) \rightarrow (\langle A' \times B, \text{snd} \rangle)$, i.e. the arrow together with the proof that it commutes with the structure maps `snd`.
- The underlying arrow in $\mathbf{A}$ is $f \times B = \text{pair}(\text{fst} \circ f, \text{snd}) : A \times B \rightarrow A' \times B$. This equals $(- \times B)(f) = \Sigma(\Delta(f))$ — that is, $\Delta(f)$ with the over-structure forgotten. Do not call the bare arrow $f \times B$ “ $\Delta(f)$ ”; it is $\Sigma(\Delta(f))$.

“ Δ separates morphisms” means Δ is faithful: $\Delta(f) = \Delta(g) \implies f = g$.

Reduction: the forgetful $\Sigma : \mathbf{A}/B \rightarrow \mathbf{A}$ is faithful — an over-morphism is determined by its underlying arrow (`OverHom.ext`). So

$$\Delta(f) = \Delta(g) \iff \Sigma(\Delta(f)) = \Sigma(\Delta(g)) \iff f \times B = g \times B,$$

and “ Δ faithful” is equivalent to “ $(- \times B)$ faithful”.

Why $(- \times B)$ is faithful when B is well-supported: from $f \times B = g \times B$, compose with the projection $\text{fst} : A' \times B \rightarrow A'$. Since $(f \times B) \circ \text{fst} = \text{fst} \circ f$ (`fst_pair`), we get $\text{fst} \circ f = \text{fst} \circ g$, where $\text{fst} : A \times B \rightarrow A$. Now B well-supported means $B \rightarrow 1$ is a cover (epimorphism onto the terminator); $\text{fst} : A \times B \rightarrow A$ is the pullback of $B \rightarrow 1$ along $A \rightarrow 1$, and covers are stable under pullback, so fst is a cover, hence epic. An epimorphism cancels on the right, so $f = g$. (In Lean: `slice_embedding_separates` in `S1_54.lean`, using `prod_fst_cover` and `cover_epi`; the wrapper `sliceEmbedFaithful` in `S1_541_RelativeCapitalization.lean` passes through `congrArg OverHom.f`, which is the Σ -faithfulness step.)

$$\begin{array}{ccc} A \times B & \xrightarrow{\Sigma(\Delta(f)) = f \times B = (- \times B)(f)} & A' \times B \\ & \searrow \text{snd} & \swarrow \text{snd} \\ & B & \end{array}$$

Figure 3: The whole triangle (arrow + commuting legs over B) is $\Delta(f)$, a morphism in $\mathbf{A}/B$; forgetting the two legs down to B (applying Σ) leaves only the top arrow $f \times B$ in $\mathbf{A}$. Δ is faithful (separates morphisms) iff this forgetting loses no information beyond what f already determines — which holds exactly when the projection $A \times B \rightarrow A$ is epic, i.e. when B is well-supported.

Set example. The underlying arrow of $\Delta(f)$ is $f \times B : A \times B \rightarrow A' \times B$, $(a, b) \mapsto (f(a), b)$.

- B nonempty (= well-supported in $\mathbf{Set}$): to recover f from $f \times B$, evaluate at (a, b) : $(f(a), b)$, so $f(a) = g(a)$ for every a — this works because some $b \in B$ exists to pair with a . Equivalently, $\text{fst} : A \times B \rightarrow A$, $(a, b) \mapsto a$, is surjective iff $B \neq \emptyset$, and surjections cancel on the right.
- $B = \emptyset$: then $A \times B = \emptyset$, so $f \times \emptyset$ and $g \times \emptyset$ are both the unique empty map $\emptyset \rightarrow \emptyset$ — equal for all f, g . Δ collapses every parallel pair. Concretely $A = \{a\}$, $A' = \{0, 1\}$, $f \equiv 0$, $g \equiv 1$: $f \neq g$ but $f \times \emptyset = g \times \emptyset$. This is exactly why the hypothesis “ B well-supported” is needed.

Where the “terminator is projective” idea belongs: $1_B : B \rightarrow B$ genuinely is the terminator of $\mathbf{A}/B$, and “the terminator is projective” is a real Freyd result (§1.525), but that concerns the category being well-pointed and is used later in the capitalization argument (§1.55). It is not what makes Δ faithful — faithfulness needs only that the projection $A \times B \rightarrow A$ is epic, which is precisely “ B well-supported”.

§3. Why $\mathbf{A} \subset \mathbf{A}/B$ is a proper inclusion (the new point)

After the final reinterpretation of §1.544, $\Delta : \mathbf{A} \rightarrow \mathbf{A}/B$ is a literal subcategory inclusion. It is a *proper* inclusion: $\mathbf{A}/B$ is strictly larger — it contains points that $\mathbf{A}$ does not. This properness is the entire engine of capitalization (§1.545–1.546): each slice step adjoins a new point.

Note. The inclusion is Δ . The forgetful functor $\Sigma : \mathbf{A}/B \rightarrow \mathbf{A}$ goes the other way; the composite $\Sigma \circ \Delta = (- \times B)$ is the round trip $\mathbf{A} \rightarrow \mathbf{A}$, not the inclusion.

§3.1. The generic point of B

In the slice $\mathbf{A}/B$ the terminator is $1_{\mathbf{A}/B} = \left(B \xrightarrow{\text{id}_B} B \right)$, and $\Delta(B) = \left(B \times B \xrightarrow{\text{snd}} B \right)$.

A *point* of $\Delta(B)$ (a global element in the slice) is a morphism $1_{\mathbf{A}/B} \rightarrow \Delta(B)$, i.e. an over-morphism from $\left(B \xrightarrow{\text{id}_B} B \right)$ to $\left(B \times B \xrightarrow{\text{snd}} B \right)$. Its underlying arrow is a map $x : B \rightarrow B \times B$ satisfying $x \circ \text{snd} = \text{id}_B$ (it commutes over B); such an x is a section of snd , of the form $x = \langle h, \text{id}_B \rangle$ for some $h : B \rightarrow B$.

The **generic point** is the diagonal $\delta = \langle \text{id}_B, \text{id}_B \rangle : B \rightarrow B \times B$ (it satisfies $\delta \circ \text{snd} = \text{id}_B$). It is a global point of $\Delta(B)$ in $\mathbf{A}/B$.

This δ generally does not come from $\mathbf{A}$. A point of B “in $\mathbf{A}$ ” is a global element $1_{\mathbf{A}} \rightarrow B$ (from $\mathbf{A}$ ’s own terminator). A well-supported B (the map $B \rightarrow 1$ is an epimorphism) need not have any such point $1 \rightarrow B$. When B has no point in $\mathbf{A}$, δ is a genuinely new point, present in $\mathbf{A}/B$ but not in $\mathbf{A}$.

This also shows Δ is faithful but not full: δ is a slice morphism between two objects that are in the image of Δ (namely $1_{\mathbf{A}/B} = \Delta(1_{\mathbf{A}})$ and $\Delta(B)$), yet δ itself is not Δ of any $\mathbf{A}$ -morphism.

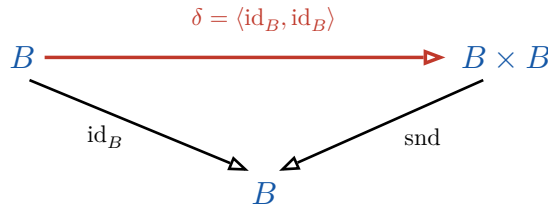


Figure 4: δ is the generic point of B , living in $\mathbf{A}/B$ but not in $\mathbf{A}$. In the $\mathbb{Z}/2$ -set example ($B = \{0, 1\}$ with swap $0 \leftrightarrow 1$), B has no point $1 \rightarrow B$ in $\mathbf{A}$ because the swap has no fixed element, yet the diagonal δ is such a point in $\mathbf{A}/B$ — the proper part of the inclusion.

§3.2. Concrete witness: a two-element swap set

Let $\mathbf{A}$ be the category of $\mathbb{Z}/2$ -sets: a $\mathbb{Z}/2$ -set is a set equipped with an involution — a self-map that is its own inverse, a “swap”. A morphism of $\mathbb{Z}/2$ -sets is a function that commutes with the swaps. (This is a small pre-regular category; in fact a topos.)

Let $B = \{0, 1\}$ with the swap $0 \leftrightarrow 1$.

B is well-supported. B is nonempty, so the unique map $B \rightarrow 1$ is onto (an epimorphism).

B has no global point in $\mathbf{A}$. A point $1 \rightarrow B$ is a morphism from the one-point trivial $\mathbb{Z}/2$ -set (the singleton with the identity swap), i.e. an element of B fixed by the swap. The swap $0 \leftrightarrow 1$ has no fixed element, so there is no point $1 \rightarrow B$ in $\mathbf{A}$.

Pass to $\mathbf{A}/B$. The diagonal $\delta = \langle \text{id}, \text{id} \rangle : B \rightarrow B \times B$ sends $b \mapsto (b, b)$; it commutes with the swap (so it is a legitimate $\mathbb{Z}/2$ -set morphism) and is a section of snd . So δ is a point of $\Delta(B)$ in $\mathbf{A}/B$.

Conclusion. δ is a point in the target $\mathbf{A}/B$ that is not in $\mathbf{A}$. It is exactly the generic point of B that §1.546 uses to capitalize B . The inclusion $\mathbf{A} \subset \mathbf{A}/B$ is therefore proper.

More points also live in $\mathbf{A}/B$ that do not come from $\mathbf{A}$: for instance, every object $(X \xrightarrow{\pi} B)$ whose domain X is not a product $B \times A$, such as a proper subobject $B' \rightarrowtail B$. But the point that matters for capitalization is the generic point δ above: slicing over B freely adjoins a global point to B .

§4. From $\mathbf{A}$ to $\underline{\mathbf{A}}$: what capitalization gains

The capitalization lemma (§1.543) gives, for every $\mathbf{A} \in \mathcal{E}$, a capital $\underline{\mathbf{A}} \in \mathcal{E}$ and a faithful representation $\mathbf{A} \rightarrow \underline{\mathbf{A}}$ in $\mathcal{E}$. The single thing gained is enough global points. Recall the two definitions this rests on:

- (§1.523) An object A is **well-pointed** if its global points $1 \rightarrow A$ cover A — equivalently, every proper subobject $A' \rightarrowtail A$ is missed by some point $1 \rightarrow A$ (there are enough points to detect every subobject).
- (§1.525) A pre-regular category is **capital** if every well-supported object is well-pointed.

§4.1. The three gains

1. **Points separate and detect.** In $\underline{\mathbf{A}}$, for every well-supported X the points $1 \rightarrow X$ jointly cover it, and every proper subobject is missed by some point. In $\mathbf{A}$ this can fail — the category may be point-poor. (Concrete case from the earlier section: in the category of $\mathbb{Z}/2$ -sets, $B = \{0, 1\}$ with swap $0 \leftrightarrow 1$ is well-supported but has no point $1 \rightarrow B$, so points saw nothing.)
2. **The terminator becomes projective** (§1.525): points lift along covers — given a cover $Y \twoheadrightarrow X$ and a point $1 \rightarrow X$, it lifts to a point $1 \rightarrow Y$. Covers are surjective on points.
3. **The global-sections functor** $\Gamma = \text{Hom}(1, -) : \underline{\mathbf{A}} \rightarrow \mathcal{S}$ ($\mathcal{S} = \text{Set}$) is a representation of pre-regular categories (§1.526): it preserves finite products, pullbacks, covers, and images. On $\mathbf{A}$ alone Γ was not even faithful — it collapsed the pointless B above to the empty set $\emptyset$.

§4.2. The payoff

Compose the embedding with global sections:

$$\mathbf{A} \hookrightarrow \underline{\mathbf{A}} \xrightarrow{\Gamma} \mathcal{S}$$

is a faithful, structure-preserving representation into $\mathbf{Set}$ (§1.55). This is the engine of Freyd’s embedding theorem: every small pre-regular (or regular) category faithfully represents into a power of $\mathbf{Set}$. In one phrase: a category is capital exactly when it has enough points to see itself in $\mathbf{Set}$ through its own global-sections functor $\Gamma = \text{Hom}(1, -)$.

$$\begin{array}{ccc} \mathbf{A} & \xrightarrow{\text{faithful; adds points}} & \underline{\mathbf{A}} \\ & & \xrightarrow{\Gamma = \text{Hom}(1, -)} \mathcal{S} (= \mathbf{Set}) \end{array}$$

composite $\mathbf{A} \rightarrow \mathcal{S}$ is faithful (§1.55)

Figure 5: Capitalization inserts $\underline{\mathbf{A}}$ so that the global-sections functor Γ becomes faithful on the enlarged category; the composite embeds $\mathbf{A}$ into $\mathbf{Set}$. In the $\mathbb{Z}/2$ -set example, $\Gamma(B) = \text{Hom}(1, B)$ = the swap-fixed elements of B , which is empty in $\mathbf{A}$ — capitalization adds the points that make Γ faithful.

§4.3. The cost

Nothing structural is lost. The embedding $\mathbf{A} \rightarrow \underline{\mathbf{A}}$ is faithful and lies in $\mathcal{E}$, so it preserves the pre-regular structure (finite products, covers, pullbacks, images). $\mathbf{A}$ sits inside $\underline{\mathbf{A}}$ intact — capitalization only adds points; it never identifies or discards anything of $\mathbf{A}$.

From $\mathbf{A}$ to $\underline{\mathbf{A}}$ we gain enough global points to turn $\Gamma = \text{Hom}(1, -)$ into a faithful representation into $\mathbf{Set}$ — converting an abstract pre-regular category into a concrete category of sets-with-structure.

§5. §1.547: choice-free, but not a terminating program

Freyd gives capitalization twice. The §1.546 construction well-orders the objects of $\mathbf{A}$ and does transfinite recursion — slice over the first well-supported object not yet capitalized; take unions at limits. This uses the axiom of choice. The §1.547 construction is choice-free: it uses the theory of rational categories [1.48]. A natural question arises: does choice-free mean we can write code that performs capitalization? The careful answer is that choice-free removes the one genuinely non-algorithmic step and makes the construction canonical and definable as data, but “choice-free” is not the same as “a terminating program that returns a finite answer”. Two things are being conflated.

§5.1. What §1.547 removes

Well-ordering an arbitrary small category is exactly the step with no algorithm — there is no procedure that well-orders a general set. §1.547 replaces it with a definite construction: the rational category [1.48] built on pairs $\langle A, F \rangle$, where $A \in \mathbf{A}$ and F is a finite set of maps from A to distinct well-supported targets. It is shown to be a relative capitalization because it is a directed union of slices. Now every object and morphism is a concrete finite description of $\mathbf{A}$ ’s own data, with no arbitrary choices. That is the content of “choice-free”: the result is canonical, hence in principle writable as data.

§5.2. Why that is still not runnable code

Two gaps remain between §1.547 and an executable program.

1. **The output is infinite.** $\underline{\mathbf{A}}$ is an ω -tower of relative capitalizations, each itself a directed union of slices, and each slice step adjoins new points. Even if $\mathbf{A}$ is finitely presented, $\underline{\mathbf{A}}$ has infinitely many objects and morphisms. So there is no finite object to compute and halt with. What you *can* write is a lazy presentation: given decidable equality, hom-membership, and composition on $\mathbf{A}$, the choice-free construction yields decidable equality, hom, and composition on $\underline{\mathbf{A}}$ together with an enumeration of its objects — a program that answers any question about $\underline{\mathbf{A}}$ on demand, not one that outputs the whole category.
2. **The input must be constructive.** The construction quantifies over predicates like “ B is well-supported”, “ $B' \rightarrow B$ is proper”, “these maps form a product diagram”. To run, these must be decidable on the presentation of $\mathbf{A}$. If they are decidable, §1.547 is genuinely computable; if not, choice-freeness alone gives no algorithm. Freyd’s argument is classical logic minus choice, which is weaker than “extracts a program” — the latter also needs decidability (or a fully constructive metatheory).

Precise statement. §1.547 makes capitalization a computable presentation of the infinite $\underline{\mathbf{A}}$, provided $\mathbf{A}$ is presented with decidable operations. It is not, and cannot be, a function returning a finite category. Choice-free means “canonical and definable as data”, not “finite” or “terminating”.

§5.3. This repo’s status

This is exactly the constructive-in-a-proof-assistant question, and the repo has not yet realized Freyd’s choice-free version. The live `capitalization_lemma` is sorry-free, but per `Freyd.lean`’s own header its axioms are `[propxt, Classical.choice, Quot.sound]` — it still uses `Classical.choice` (it went through a well-ordering / uniform-successor route with `Zermelo` built from `Classical.choice`, in `WellOrdering.lean`). Genuinely eliminating `Classical.choice` here — implementing the §1.547 rational-category route as pure data — is precisely what would turn the Lean development into code you could compute with. That is an open task in the codebase, not something already done.

aspect	§1.546 (choice)	§1.547 (choice-free)
key step	well-order the objects; transfinite recursion	rational category on pairs $\langle A, F \rangle$; directed union of slices
uses AC?	yes (well-ordering)	no
canonical?	no (depends on the chosen well-ordering)	yes (determined by $\mathbf{A}$ ’s data)
length	transfinite (not, in general, countable)	ω -tower of directed unions
code?	no algorithm (cannot well-order a general set)	computable lazy presentation if $\mathbf{A}$ ’s operations are decidable

Table 1: The choice-free §1.547 route is what could, in principle, be turned into code; the §1.546 route cannot, because well-ordering a general set is not algorithmic.

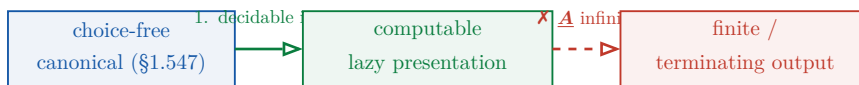


Figure 6: Choice-free gives a canonical construction; adding decidable operations on $\mathbf{A}$ upgrades it to a computable, on-demand presentation of $\underline{\mathbf{A}}$; but it never becomes a finite or terminating output, because $\underline{\mathbf{A}}$ is infinite.